

Wigner Resolution from Action Capacity

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(Dated: 2026-06-17)

Smoothing the Wigner function gives a non-negative phase-space portrait of a quantum state and the scale of that smoothing has been chosen rather than derived. We show that the scale is fixed by the state. The state's covariance defines an elliptical Heisenberg cell of action capacity $\pi \Delta x \Delta p$, and the cell admits a family of de Gosson quantum blobs deforming at constant action $\hbar/2$. Convolving the Wigner function to a single $\hbar/2$ blob renders it non-negative. Integrating over the family reaches the resolution $\pi \hbar^2 / (\Delta x \Delta p)$, the symplectic polar dual of the Heisenberg cell. The portrait at this resolution is everywhere non-negative and holds the structure of the Wigner function in place. Wigner resolution is set by the state's action capacity alone, sharpens as that capacity grows, and carries no free parameter.

The Wigner function represents a quantum state in phase space [1, 2]. It is defined pointwise below Planck's quantum of action, where Hudson's theorem forces it negative [3]. Its negativity is a resource for quantum advantage [4, 5] and a measure of nonclassicality [6]. A non-negative distribution supports classical sampling [7–11] and follows from smoothing the Wigner function at a scale chosen rather than derived. Here the state's action capacity $\pi \Delta x \Delta p$ sets that scale without a free parameter.

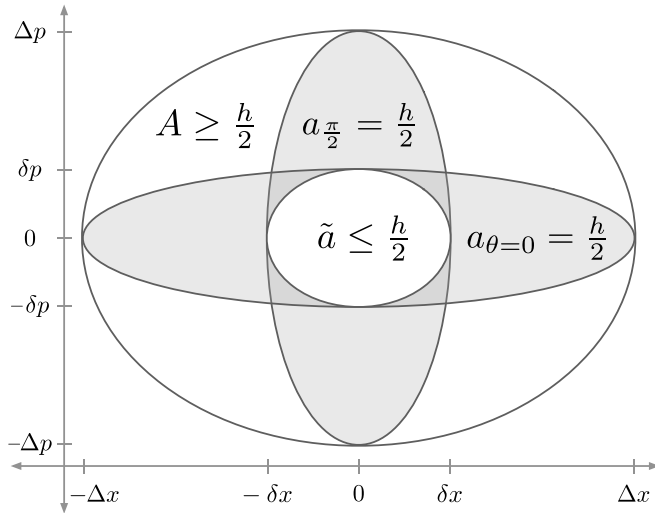


FIG. 1. Heisenberg cell, quantum blobs at principal axes, and quorum cell. The Heisenberg cell is the state's covariance ellipsoid with action capacity $A = \pi \Delta x \Delta p \geq \hbar/2$. The quantum blob family (Fig. 2) has two members at the principal axes, $a_{\theta=0} = \pi \Delta x \delta p = \hbar/2$ and $a_{\pi/2} = \pi \delta x \Delta p = \hbar/2$. Zurek's relations $\delta p \sim \hbar/\Delta x$ and $\delta x \sim \hbar/\Delta p$ locate the semi-axis scales from the state's action capacity. The quorum cell $\tilde{a}$ is an ellipse with semi-axes $\delta x, \delta p$ and action $\tilde{a} = \pi \delta x \delta p = \pi \hbar^2 / (\Delta x \Delta p)$. A circumscribes $a_{\theta=0}$ and $a_{\pi/2}$, which circumscribe $\tilde{a}$.

Planck divided phase space into elementary regions of action [12]. Heisenberg's uncertainty principle $\sigma_x \sigma_p \geq \hbar/2$ [13–15] fixes their minimum area. Geometrically, de

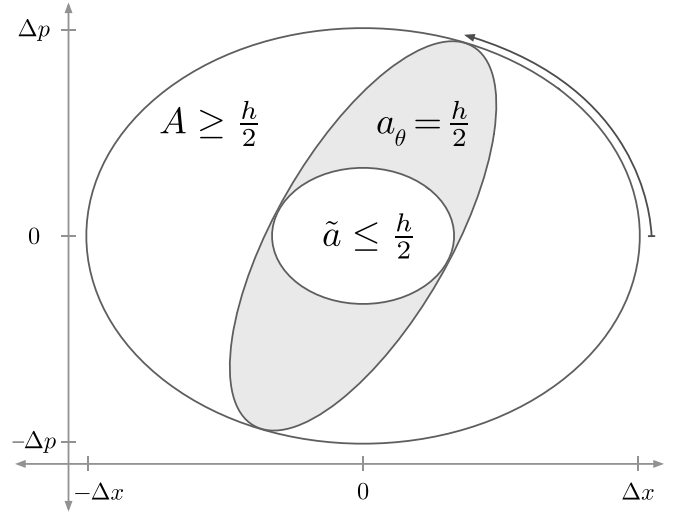


FIG. 2. Heisenberg cell, quadrature family of quantum blobs, and quorum cell. The Heisenberg cell A circumscribes the quantum blob family $\{a_\theta\}$, which circumscribes the quorum cell $\tilde{a}$. Across θ the family deforms at constant action $a_\theta = \hbar/2$. At the principal angles $\theta = 0$ and $\theta = \pi/2$, the family recovers $a_{\theta=0}$ and $a_{\pi/2}$ from Fig. 1.

Gosson gave this action region the form of the state's covariance ellipse, with symplectic capacity $\pi \Delta x \Delta p$ [16] and semi-axes $\Delta x = \sqrt{2} \sigma_x$ and $\Delta p = \sqrt{2} \sigma_p$. This is the Heisenberg cell (Figs. 1, 2) of action capacity,

$$A = \pi \Delta x \Delta p \geq \hbar/2. \quad (1)$$

The Wigner function carries structure at small scales. Zurek located those scales, $\delta p \sim \hbar/\Delta x$ and $\delta x \sim \hbar/\Delta p$, from the state's action capacity [17]. We identify them as the principal axes of a family of de Gosson quantum blobs (Fig. 1),

$$a_{\theta=0} = \pi \Delta x \delta p = \hbar/2, \quad a_{\pi/2} = \pi \delta x \Delta p = \hbar/2. \quad (2)$$

The capacity $\Delta x, \Delta p$ runs along the major axes, the resolution $\delta x, \delta p$ along the minor.

Between these principal members the quantum blob family $\{a_\theta\}$ deforms across $\theta \in [0, \pi)$ at constant action

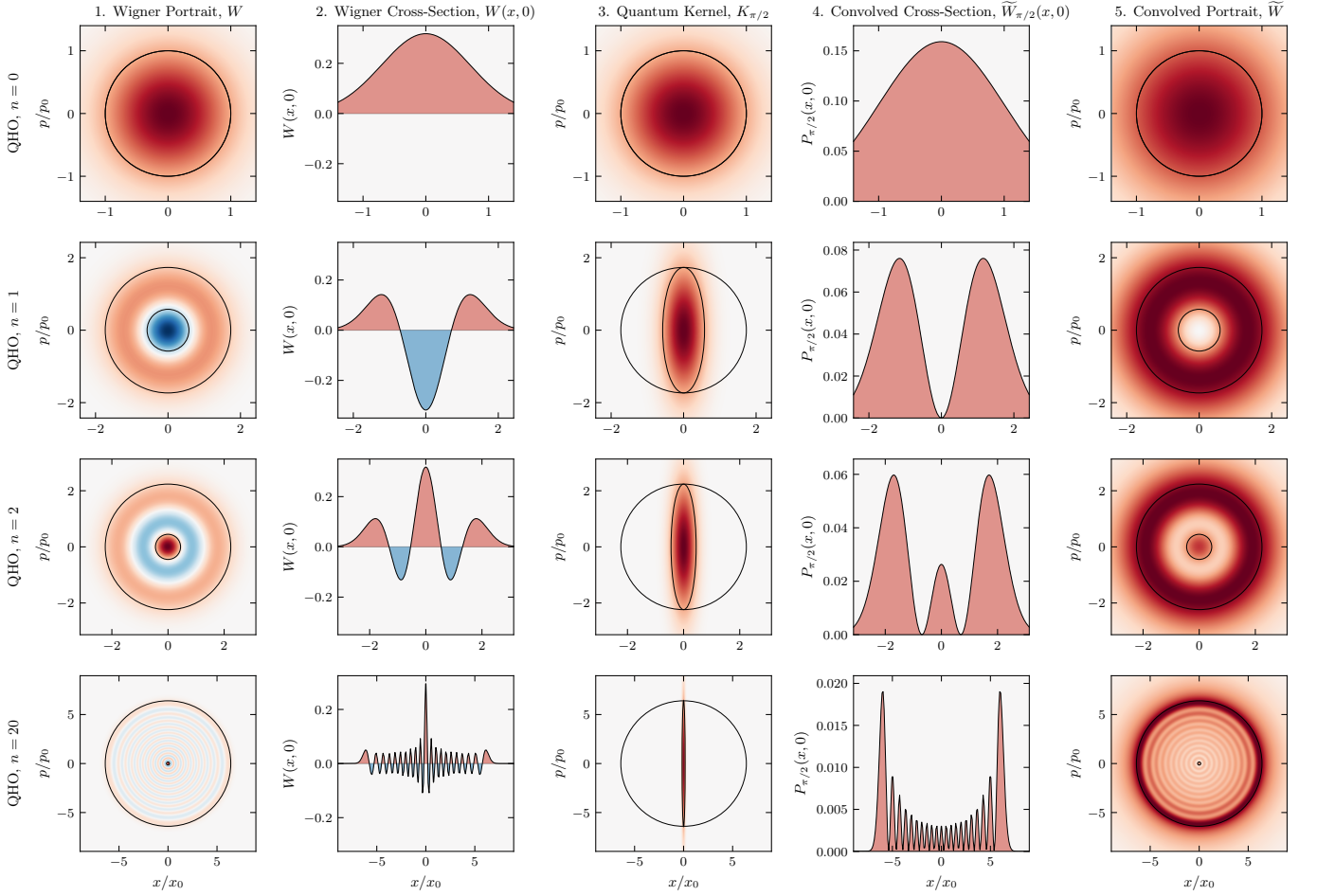


FIG. 3. **Harmonic oscillator.** Rows correspond to Fock states $n = 0, 1, 2, 20$ with action capacity $A/(\hbar/2) = 2n + 1$. Column 1: Wigner function $W(x, p)$ (negativity in blue) overlaid with the Heisenberg cell A and quorum cell $\tilde{a}$. Centered at the covariance origin, $\tilde{a}$ marks the resolution of the finest structure (the Zurek scale) of the bare, unconvolved Wigner function. Column 2: Cross-section $W(x, 0)$ exhibiting negative dips. Column 3: Quantum kernel $K_{\pi/2}$ overlaid with A and the quantum blob $a_{\pi/2}$ (action $\hbar/2$, restoring resolution up to Planck's quantum of action). Column 4: Convolved cross-section $\tilde{W}_{\pi/2}(x, 0)$ at quadrature angle $\theta = \pi/2$, which is non-negative. Column 5: The portrait $\tilde{W}(x, p)$, integrated over the blob family across all θ , overlaid with A and $\tilde{a}$ (symplectic polar duals). It is everywhere non-negative, with positive lobes at the positions of those in W , resolved at $\tilde{a}$. For the vacuum ($n = 0$), A and $\tilde{a}$ coincide as a Gaussian blob at $A = \hbar/2$. As n increases, A grows and $\tilde{a}$ reciprocally shrinks ($\tilde{a} A = (\hbar/2)^2$). At $n = 20$, the rings condense toward a classical energy trajectory, leaving $\tilde{a}$ as a fine central cell. Axes are in the oscillator scales $x_0 = \sqrt{\hbar/m\omega}$ and $p_0 = \sqrt{\hbar m\omega}$.

$\hbar/2$ (Fig. 2). Each blob a_θ is the preimage of the rigid ellipse $\hat{a}_\theta$ under the scaling $X = x/\Delta x$, $P = p/\Delta p$,

$$a_\theta = \{(x, p) : (X, P) \in \hat{a}_\theta\}. \quad (3)$$

The rotated quadratures $X_\theta = X \cos \theta + P \sin \theta$ and $P_\theta = -X \sin \theta + P \cos \theta$ are the homodyne quadratures of optical tomography at phase θ . In the scaled coordinates $\hat{a}_\theta$ is the rotation by θ of a single ellipse,

$$\hat{a}_\theta = \left\{ (X, P) : X_\theta^2 + \frac{A^2}{(\pi\hbar)^2} P_\theta^2 \leq 1 \right\}, \quad (4)$$

inscribed in the unit disk at every θ , so the Heisenberg cell A circumscribes each blob a_θ .

The Wigner function of the state is

$$W(x, p) = \frac{1}{\pi\hbar} \int \psi^*(x+s) \psi(x-s) e^{2ips/\hbar} ds. \quad (5)$$

The Wigner function of each quantum blob a_θ is the centered Gaussian whose $\hbar/2$ ellipse is a_θ ,

$$K_\theta(x, p) = \frac{1}{\pi\hbar} \exp \left[-X_\theta^2 - \frac{A^2}{(\pi\hbar)^2} P_\theta^2 \right]. \quad (6)$$

The convolution of W with each quantum kernel K_θ is

$$\tilde{W}_\theta(x, p) = (W * K_\theta)(x, p). \quad (7)$$

Convolving W with the kernel K_θ of a single quantum blob a_θ lifts it up to Planck's quantum of action and yields a non-negative function [8, 9, 18].

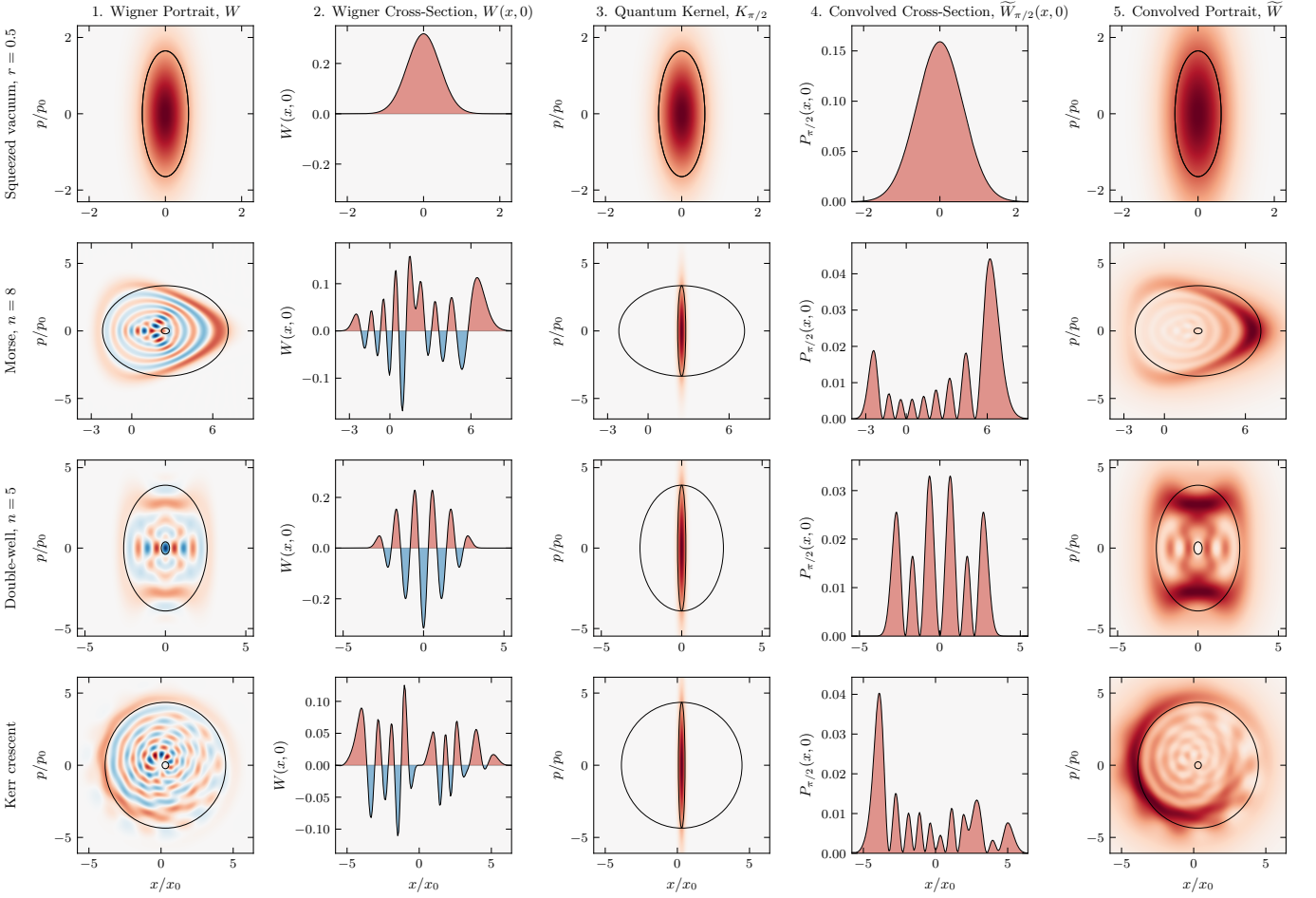


FIG. 4. **Squeezed vacuum, eigenstates, and the Kerr crescent.** Rows correspond to the squeezed vacuum ($r = 0.5$), the Morse eigenstate ($n = 8$), the double-well eigenstate ($n = 5$), and the Kerr crescent, a coherent state sheared at constant photon number. Columns are as in Fig. 3. The Kerr state is shown in the principal frame of its covariance, where the cell overlays are axis-aligned. The portrait $\widetilde{W}$ (Column 5) remains strictly non-negative across all four states, resolved at $\tilde{a}$.

Zurek also located the scale $a \sim \hbar \times (\hbar/A)$ [17, 19] of the chessboard structure in the compass state. We identify its rotationally invariant form as the quorum cell $\tilde{a}$ (Fig. 5, column 1). Named after the quorum of observables in optical tomography [20], $\tilde{a}$ is the ellipse with semi-axes $\delta x, \delta p$ (Figs. 1, 2) and action

$$\tilde{a} = \pi \delta x \delta p = \frac{\pi \hbar^2}{\Delta x \Delta p}. \quad (8)$$

$\tilde{a}$ is the resolution of the state's finest structure in the bare Wigner function (Figs. 3, 4, 5, column 1). The quorum cell $\tilde{a}$ is the symplectic polar dual of the Heisenberg cell A [21, 22]. The duality is reflexive, so A and $\tilde{a}$ each fix the other, and antimonotone, so $\tilde{a}$ shrinks as A grows. Each blob a_θ is circumscribed by A and circumscribes $\tilde{a}$, nesting as $A \supseteq a_\theta \supseteq \tilde{a}$ (Fig. 2). A quorum of these a_θ blobs, each respecting $\hbar/2$, reaches $\tilde{a}$ together. Resolution and capacity are reciprocal at the quantum of action, $\tilde{a} A = (\hbar/2)^2$.

A single blob commits to a quadrature the state does

not privilege. Integrating $\widetilde{W}_\theta$ over the family privileges none, giving the phase-space portrait of the state at the resolution $\tilde{a}$,

$$\widetilde{W}(x, p) = \frac{1}{\pi} \int_0^\pi \widetilde{W}_\theta(x, p) d\theta. \quad (9)$$

$\widetilde{W}_\theta \geq 0$ for every θ , so $\widetilde{W} \geq 0$ everywhere. Where tomographic reconstruction inverts marginals pointwise to recover W [23–25], convolving W with the family $\{K_\theta\}$ over the same θ builds the non-negative portrait $\widetilde{W}$.

A quantum blob a_θ too large for A to circumscribe, or too small to circumscribe $\tilde{a}$, is a blob of another state and portrays that state, not this one [21, 26]. Because every $\hbar/2$ blob a_θ of this state circumscribes the quorum cell $\tilde{a}$, and a portrait finer than $\tilde{a}$ everywhere would smooth W below the $\hbar/2$ action non-negativity requires [8, 9], $\tilde{a}$ is the finest resolution any non-negative portrait attains, and the family $\{a_\theta\}$ reaches it across $\theta \in [0, \pi)$.

A fixed vacuum-width Gaussian [7] smooths every state by the same amount, matching only the vacuum

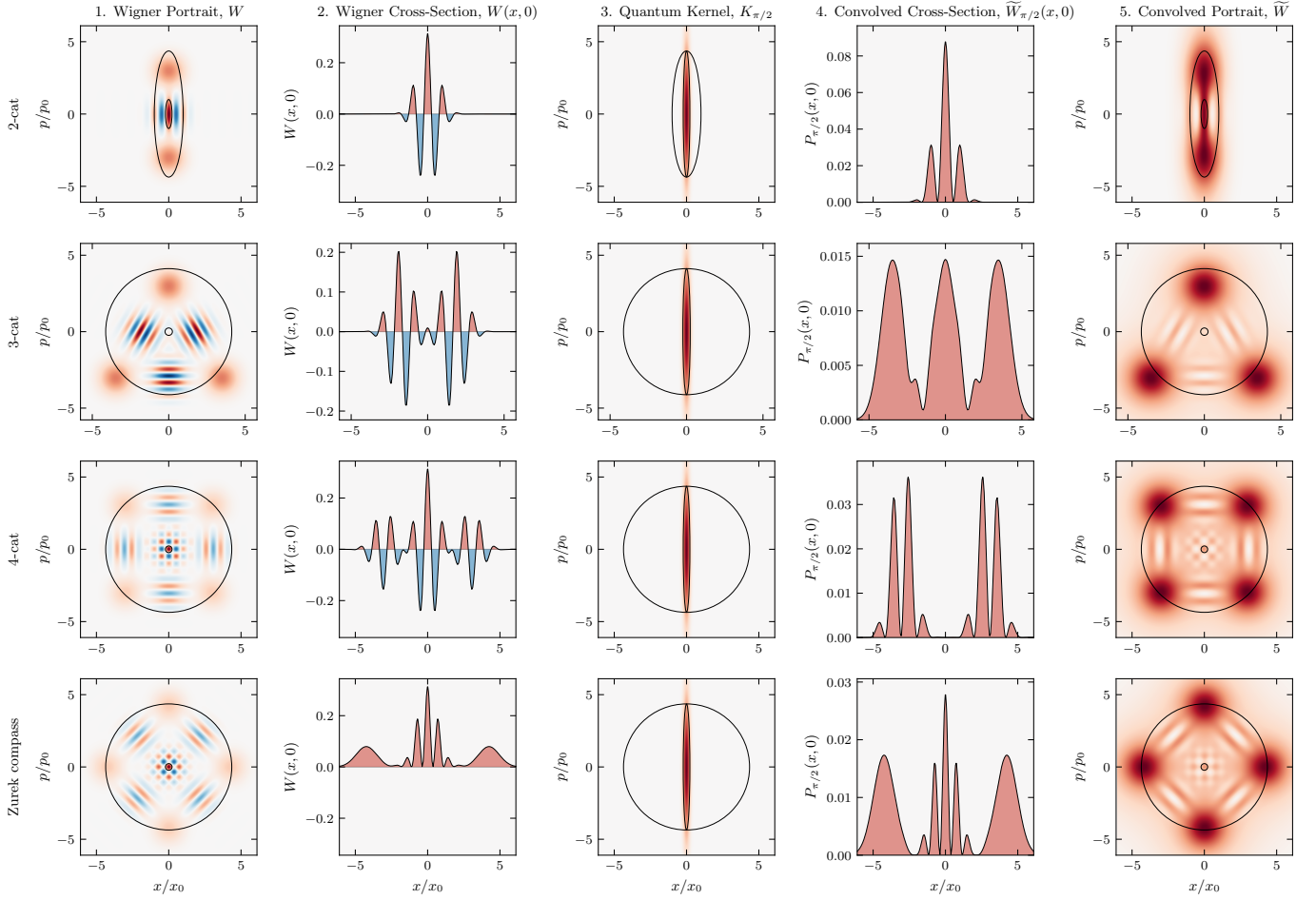


FIG. 5. **Cat states.** Rows correspond to the 2-cat, 3-cat, 4-cat, and Zurek compass states. Columns are as in Fig. 3. As before, the portrait $\tilde{W}$ (Column 5) is everywhere non-negative, with positive lobes at the positions of those in W , resolved at $\tilde{a}$. The 4-cat and Zurek compass states differ by a $\pi/4$ rotation in phase space.

$A = h/2$ and erasing the fine structure of larger states $A > h/2$. Methods with a free parameter [27–29] resolve more finely, the scale chosen rather than derived. The scale here is derived. The state’s action capacity A fixes its quorum cell $\tilde{a}$ as the symplectic polar dual, and the family $\{a_\theta\}$ reaches $\tilde{a}$ with no quantity left free.

The principal frame, where the Heisenberg cell A is axis-aligned, costs no generality. The Wigner function transforms covariantly under the symplectic group [18], so the map that diagonalizes a state’s covariance carries it to this frame at preserved action. The Kerr crescent, a sheared coherent state, is resolved this way (Fig. 4). In one degree of freedom, A fixes the family uniquely, and the sub-Planck structure is already present for states with $A > h/2$. In higher dimensions, action $h/2$ no longer singles out a blob [26], symplectic capacity and volume diverge, and the higher-dimensional case remains for future work.

Figures 3, 4, and 5 show numerical results across twelve states. The portrait is non-negative to machine precision, and the cross-spectral phase between W and $\tilde{W}$ vanishes,

confirming that the smoothing displaces no feature [30].

The uncertainty product $\sigma_x \sigma_p$ grows unbounded, the micro-to-macro transition Heisenberg [31] and Bohr [32] identified as requiring special care. The quantum blob does not. As A grows, the family $\{a_\theta\}$ deforms at fixed action $h/2$. Squeezing reaches the same resolution by external apparatus [33], while here the deformation is set by the state. A Schrödinger cat with capacity $\Delta x \sim 10$ cm and $\Delta p \sim 1$ kg·m/s carries structure in its Wigner function at $\delta x = \hbar/\Delta p \sim 10^{-34}$ m [34]. The correspondence principle follows from $A \rightarrow \infty$, the physical limit underlying the heuristic $\hbar \rightarrow 0$.

The scale called sub-Planck [17] is the minor axis δ of an $h/2$ blob, narrowing as the conjugate capacity Δ grows. The negative Wigner function W and its non-negative portrait $\tilde{W}$ have their structure at a single resolution $\tilde{a}$, set by the state’s action capacity A . Heisenberg’s product gives the capacity. The Zurek scales are the axes of de Gosson quantum blobs, and give the resolution. As the action capacity grows to $\pi \Delta x \Delta p \geq h/2$ the resolution sharpens to $\pi \hbar^2/(\Delta x \Delta p) \leq h/2$. We read

this structure as the geometry of phase space itself, which de Gosson takes as prior to the uncertainty principle [35].

The author thanks M. de Gosson for helpful comments. This research received no external funding. Numerical computations were performed with QuTiP [36] and SciPy [37]. The code reproducing the numerical results, together with Supplemental Material mapping this Letter onto Zurek and de Gosson, is openly available [30].

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